Program 3

Step 1: Create App.py and requirements.txt

```
1rv24mc024_chandan_athreya_s@JARVIS:~/Devops-LabSessions/prog3$ cat requirements.txt
Flask == 2.3.3
1rv24mc024_chandan_athreya_s@JARVIS:~/Devops-LabSessions/prog3$ cat app.py
from flask import Flask

app = Flask(__name__)

@app.route("/")

def run():
        return "Hello from Dockerized Flask Application"

if __name__ == "__main__":
        app.run(host="0.0.0.0", port=5000)
```

Step 2: Create Docker file

```
1rv24mc024_chandan_athreya_s@JARVIS:~/Devops-LabSessions/prog3$ cat Dockerfile
#Create a python image by pulling latest code
FROM python:latest

#Initialize Working Directory
WORKDIR /app


COPY requirements.txt .

#Install Dependencies
RUN pip install --no-cache-dir -r requirements.txt

#Copy the Source file
COPY . .

#Expose the Port 5000
EXPOSE 5000

#Run the app file
CMD ["python", "app.py"]
```

Step 3: Create Docker image

```
1rv24mc024_chandan_athreya_s@JARVIS:~/Devops-LabSessions/prog3$ docker build -t program-3 .
[+] Building 2.2s (10/10) FINISHED
 => [internal] load build definition from Dockerfile
 => => transferring dockerfile: 359B
 => [internal] load metadata for docker.io/library/python:latest
 => [internal] load .dockerignore
 => => transferring context: 2B
 => [1/5] FROM docker.io/library/python:latest@sha256:1ad1a43b5e2478e62056bbc28028afd858185d73bf4d6a439cbb058b6800a96d
 => [internal] load build context
 => => transferring context: 295B
 => CACHED [2/5] WORKDIR /app
 => CACHED [3/5] COPY requirements.txt .
 => CACHED [4/5] RUN pip install --no-cache-dir -r requirements.txt
 => [5/5] COPY . .
 => exporting to image
 => => exporting layers
 => => writing image sha256:76fa08070ad21e8d6a3430c0b75ffeabc35d347799eea2705a31f10cb01ab90f
 => => naming to docker.io/library/program-3
```

Step 4: Run Docker image as a container

```
1rv24mc024_chandan_athreya_s@JARVIS:~/Devops-LabSessions/prog3$ docker run -p 5000:5000 program-3
 * Serving Flask app 'app'
 * Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
 * Running on all addresses (0.0.0.0)
 * Running on http://127.0.0.1:5000
 * Running on http://172.18.0.2:5000
Press CTRL+C to quit
172.18.0.1 - - [07/Nov/2025 04:08:02] "GET / HTTP/1.1" 200 -
```

Step 5: Output